Red Bird Challenge Camp Portfolio

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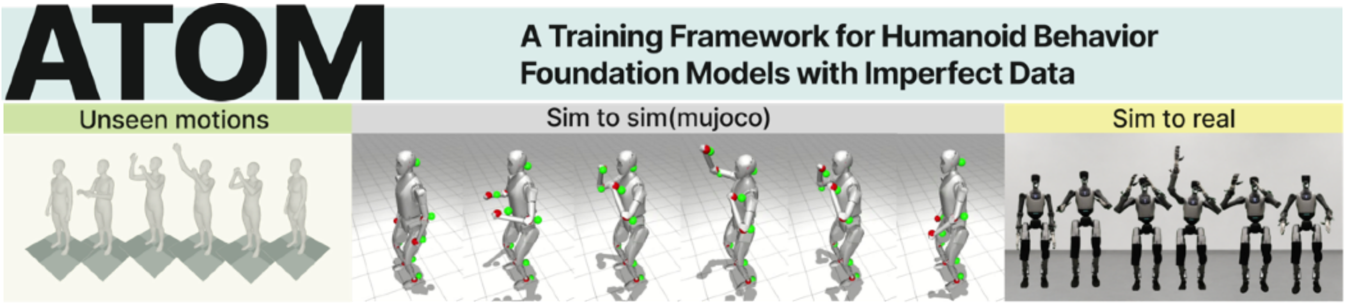
Major: Computer Science and Technology

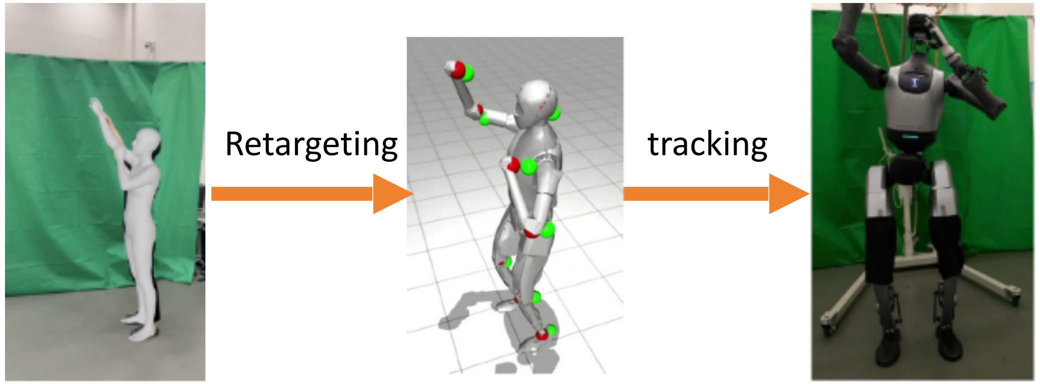
Robotics · Embodied AI · Intelligent Systems

Experienced in integrating perception, control, embedded systems, and reinforcement learning into real-world deployment.

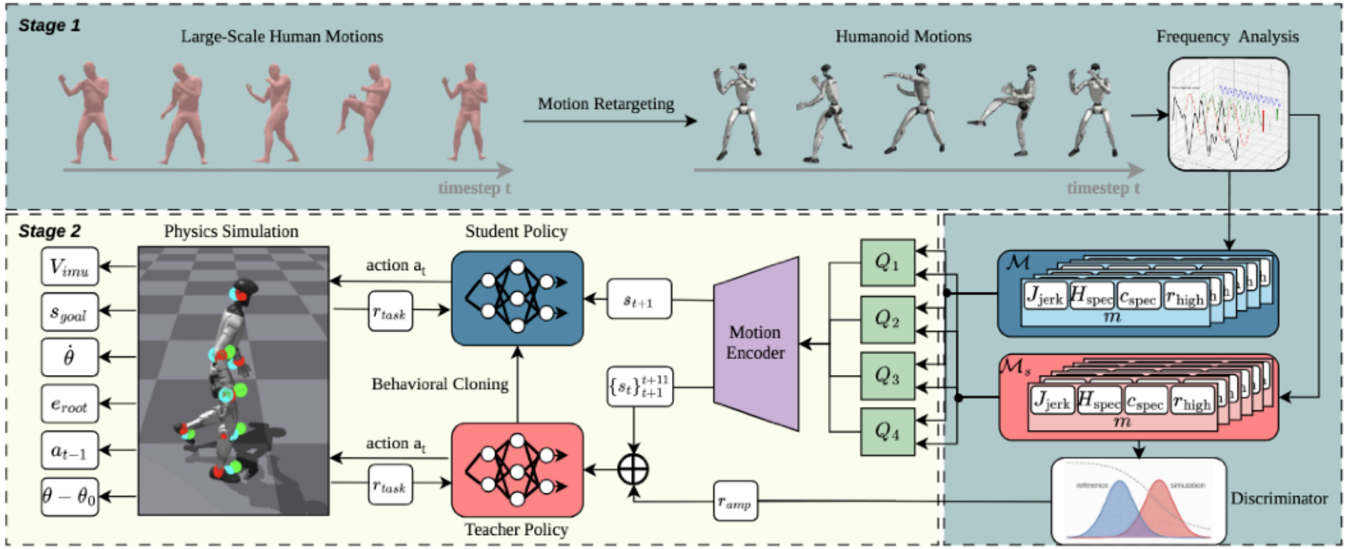
Humanoid Whole–Body Control

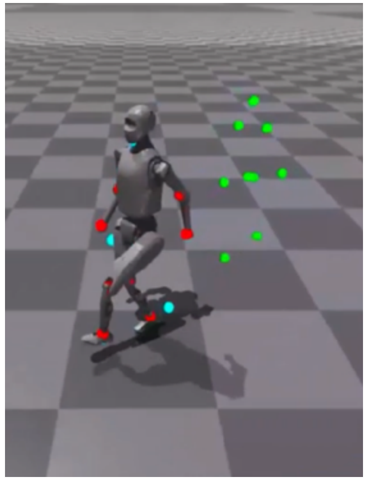


Our framework extracts human skeletal motion from video, retargets it to the humanoid robot, and trains RL policies to reproduce movements while ensuring stability and safety



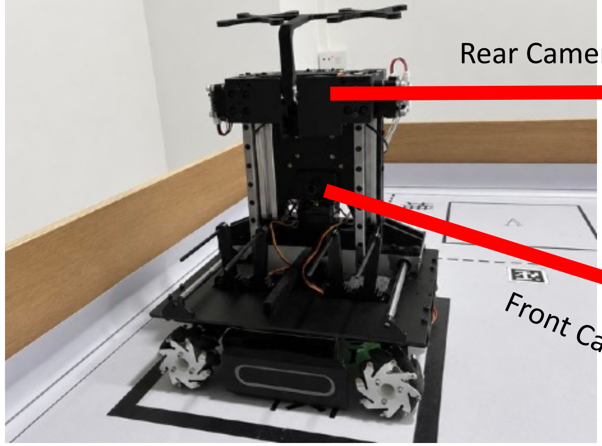
Human motion video   Robot motion data   Real world tracking



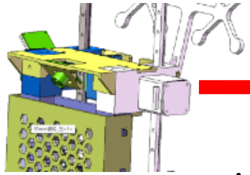


STEM Education Robot Kit

Robot Pose + Planned Path



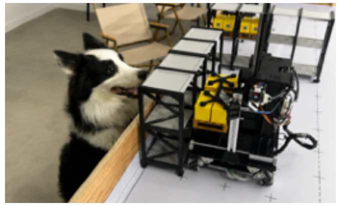
Rear Camera



Ground-based
AprilTag Localization

Front Camera

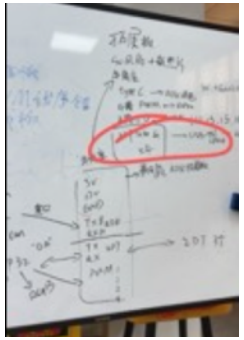
AprilTag Alignment



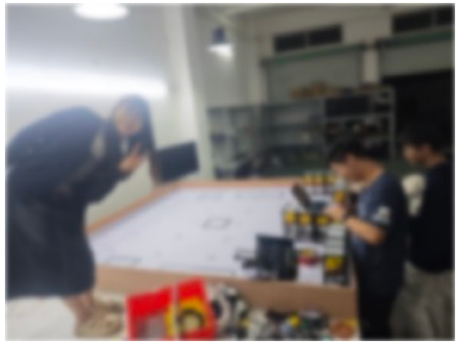
Pallet Pickup

Placement

Collaborated on hardware–software integration to transform the robot from a prototype into a **mass-produced educational product**, later **deployed and sold to student users**.

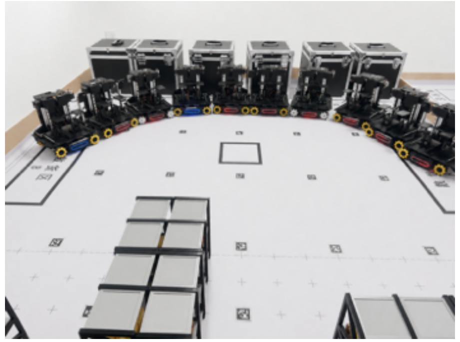


Team Brainstorming
for Robot Design

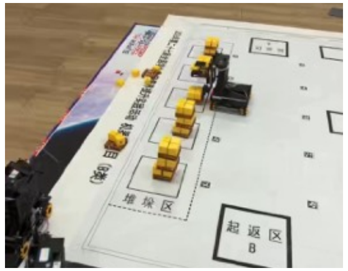


Teamwork in Robot
Development

Robot Assembly

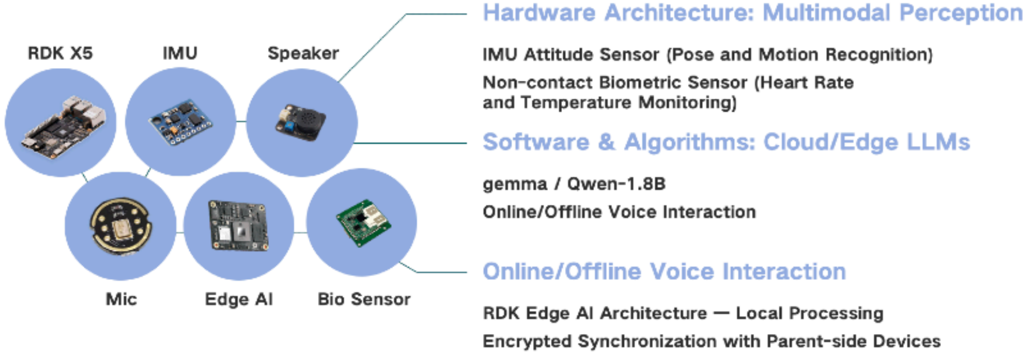


Batch Production of Our Robots



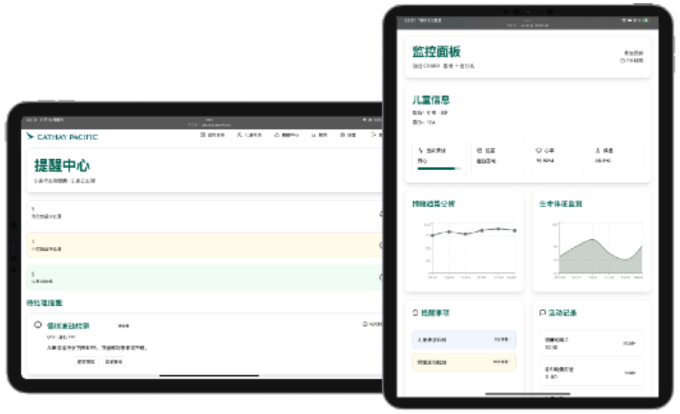
From Prototype to Mass-Produced Educational Robot

Cathay Hackathon — AI Companion for Unaccompanied Children



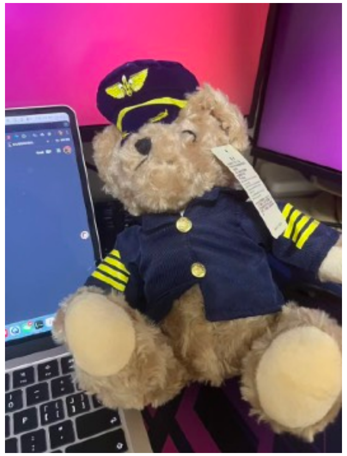
AI System Architecture

Built an edge-AI companion prototype for unaccompanied children during air travel. My work focused on the **edge AI** architecture and **LLM interaction** pipeline, supporting multimodal sensing, online/offline voice interaction, and device–cloud communication



UI Design

Interdisciplinary Team Collaboration



Edge AI Companion Prototype

AI-Assisted Child Companion Scenario

ICRA 2024 Sim2Real Challenge

Task Setup

Simulation

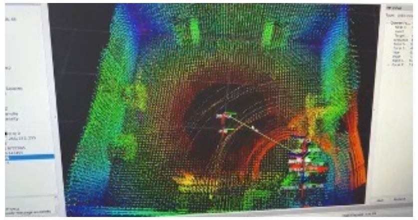
Object Detection

On-Site Real-World Deployment at ICRA

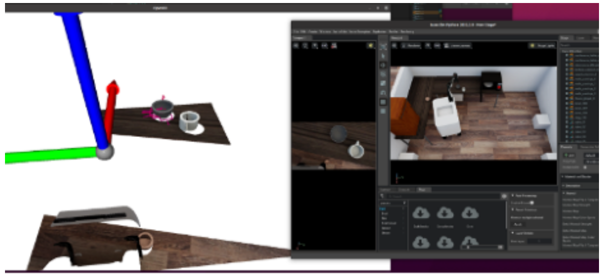
Developed an integrated **perception and control pipeline** for a mobile manipulator, combining **SLAM-based navigation**, **object detection**, **grasp planning**, and **robotic manipulation** for autonomous task execution across **simulation** and **real-world** environments.

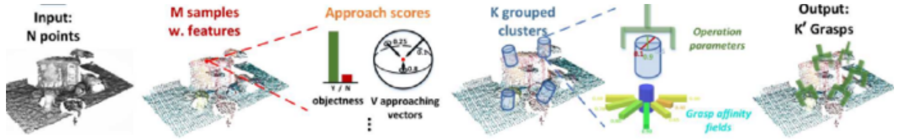
Technical Contributions

- SLAM-based localization and autonomous navigation

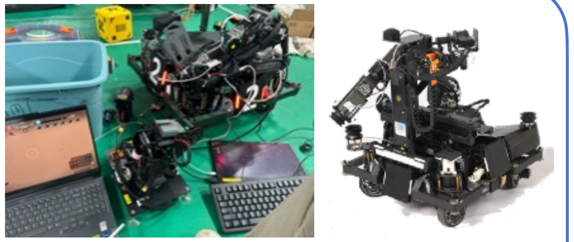


- Grasp pose prediction and robotic manipulation

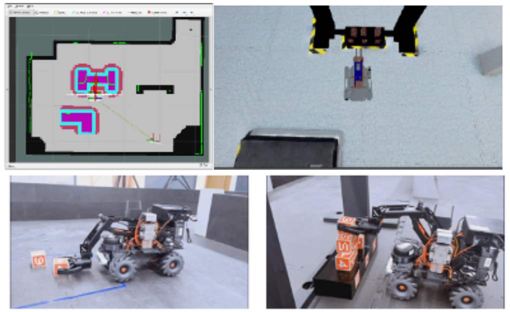




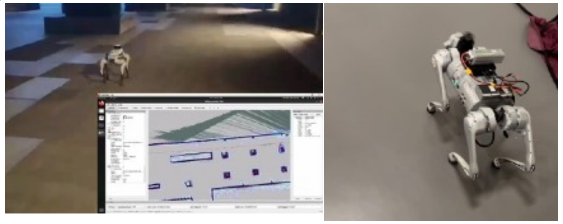
Other Projects & Awards



ROBOMASTER Engineering Robot



Mineral Grasping Robot for Sim2real Challenge



Quadruped SLAM Navigation

Engineering Practice & Collaboration